Knee Surg, Sports Traumatol, Arthrosc
(2001) 9 : 86–91

KNEE

DOI 10.1007/s001670100191

K. E. Webster
J. A. Feller
K. A. Hameister

Bone tunnel enlargement following anterior cruciate ligament reconstruction: a randomised comparison of hamstring and patellar tendon grafts with 2-year follow-up

Received: 7 August 2000
Accepted: 5 December 2000
Published online: 27 February 2001

K. E. Webster · J. A. Feller (✉)
La Trobe University
Musculoskeletal Research Centre,
Victoria 3086, Australia
e-mail: jfeller@lumc.com.au,
Tel.: +61-3-94738850,
Fax: +61-3-94738857

K. A. Hameister
La Trobe University Medical Centre,
Bundoora 3083, Australia

Abstract Radiographic tibial and femoral bone tunnel enlargement has been demonstrated following anterior cruciate ligament (ACL) reconstruction. This study investigated whether bone tunnel enlargement differs between four-strand hamstring (HS) and patellar tendon (PT) ACL reconstructions over the course of a 2-year follow-up. Patients undergoing primary ACL reconstruction (n=65) were randomised to receive either a PT or HS autograft. Femoral fixation in both groups was by means of an Endobutton. On the tibial side the PT grafts were fixed using a metallic interference screw, and the HS tendons by sutures tied to a fixation post. The PT grafts were inserted such that the proximal end of the distal bone block was within 10 mm of the tibial articular surface, resulting in a portion of free patellar tendon in the femoral tunnel immediately proximal to the articular surface. Patients were reviewed after 4 months and 1 and 2 years. Tunnel enlargement was determined by measuring the widths of the femoral and tibial tunnels with a digital caliper in both lateral and anteroposterior radiographs. Because of the presence of the interference screw and the proximity of the bone block to the tibial articular surface, the tibial tunnel could not be reliably measured in the PT group. Measurements were corrected for magnification, and changes in tunnel width were recorded relative to the diameters drilled at surgery. Standard clinical measures were also noted. In 32% of patients in the PT group there was femoral tunnel obliteration from 4 months onwards. For the other patients there was a significantly greater increase in femoral tunnel width in the HS group than in the PT group at each follow-up, but no significant change with time. There was also a marked increase in tibial tunnel width in the HS group at 4 months but not thereafter. There was no relationship between tunnel enlargement and clinical measurements. Although tunnel enlargement is more common and greater with HS grafts, it does not appear to affect the clinical outcome in the first 2 postoperative years. Femoral suspensory fixation does not in itself appear to be the principal cause of femoral tunnel enlargement, at least for PT grafts.

Keywords Anterior cruciate ligament reconstruction · Bone tunnel enlargement · Hamstring tendon graft · Patellar tendon graft

Introduction

Tibial and femoral bone tunnel enlargement following anterior cruciate ligament (ACL) reconstruction has been observed on postoperative radiographs. There are now an increasing number of studies that have reported this phenomenon with a variety of surgical techniques [1, 2, 3, 5, 8, 9, 10, 13, 19]. Despite the number of reports of this phenomenon, its cause and clinical relevance remain unclear. This may in part be due to the fact that many of the studies have been retrospective with a limited ability to control for potentially confounding variables.

Various potential causes of bone tunnel enlargement have been suggested, with a common distinction made between biological and mechanical factors (see [6] for review). Suspensory fixation methods are associated with greater distances between the articular surface and the point of graft fixation than are other methods of graft fixation. Greater motion at the graft tunnel interface has been ascribed to the elastic nature of such constructs – the so-called bungy effect – and has been has been suggested as a cause of bone tunnel enlargement [5, 7, 10, 13, 19].

Only two studies have compared bone tunnel enlargement between hamstring (HS) and patellar tendon (PT) autografts [3, 10]. Both of these showed significantly greater tunnel enlargement with the use of HS grafts, but they came to different conclusions regarding the cause of the enlargement process. L'Insalata et al. [10] argued for a mechanical cause, due to the fixation points of the HS graft being a greater distance from the articular surface than the fixation points of the PT graft. However, Clatworthy et al. [3] found no enlargement in the femoral tunnel of the PT group despite the use of suspensory fixation. Marked enlargement was, however, observed in the HS group with the same method of femoral fixation. The authors concluded that graft fixation has only a subtle effect on tunnel enlargement. This is supported by a more recent report that showed tunnel enlargement with both elastic and rigid fixation methods of HS grafts [2].

The two studies comparing PT and HS grafts [3, 10] were retrospective and non-randomised. Consequently the duration of follow-up was significantly different between the two graft types and the ability to account for confounding variables was limited. In both studies follow-up was on one occasion only. This precluded comment on the timing of the tunnel enlargement process. In the present study a prospective randomised design was employed to (a) evaluate radiographic bone tunnel enlargement at different time points and (b) compare HS and PT grafts using the same method of suspensory fixation on the femoral side.

Patients and methods

The inclusion criteria for the study were: age 18–40 years, ACL ligament rupture having occurred more than 3 weeks and less than 12 months prior to surgery, no previous cruciate ligament damage to either knee, no previous surgery to the affected knee, no collateral ligament injury of greater than grade II severity, no radiographic evidence of osteoarthritis, no chondral disruption of greater than Noyes et al. [14] grade IIA or grade IIA greater than 1 cm in diameter, and no meniscal requiring repair. Stable meniscal tears and tears treated by partial meniscectomy were not exclusion criteria.

Table 1 General data on the study subjects

	Hamstring graft (n=33)	Patellar tendon graft (n=28)	p
Age (years)	27±6.1	26±6.3	NS
Males:females	23:10	20:8	NS
Interval injury–surgery	10/52	13/52	NS
Preoperative sports activity level (0–100)[a]	87±13	92±8	NS
Preoperative occupational rating (0–60)[b]	22±12	23±14	NS

[a] Noyes et al. [16]
[b] Noyes et al. [17]

Sixty-five patients undergoing primary ACL reconstruction who met the inclusion criteria were randomised to receive either a central third bone patellar tendon bone autograft or a four-strand (doubled semitendinosus/doubled gracilis) hamstring autograft. Patients were randomised after the initial arthroscopy at the commencement of the ACL reconstructive surgery by means of a computer-generated list of random numbers. All patients were informed about the nature of the experiment and gave their consent. Four patients discontinued the study (three drop out, one early graft rupture). General data on the remaining 61 patients are shown in Table 1.

The two groups were similar in terms of age, sex, interval between injury and surgery, preoperative sports activity level and preoperative occupational rating (Table 1). At operation there was no difference between the two groups in terms of the presence or severity of concomitant meniscal and chondral pathology.

Surgical technique

All patients underwent an arthroscopically assisted single incision ACL reconstruction performed by a single surgeon. Graft constructs were statically pretensioned at 20 lb (89 N) for 5 min prior to insertion. In both groups femoral fixation was by means of an Endobutton (Smith and Nephew Endoscopy, Mansfield, Mass., USA) attached to the graft with a doubled 3-mm polyester tape. The distal ends of the PT grafts were fixed with a cannulated metallic interference screw. The distal ends of the HS grafts were fixed by means of an Acufex fixation post (Smith and Nephew Endoscopy) to which the two ends of a #5 Ethibond (Ethicon, Somerville, N.J., USA) whip stitch in each tendon were tied. The posts were inserted into the tibia distal to the tunnel mouth.

The PT grafts were inserted such that the proximal end of the distal bone block was within 10 mm of the tibial articular surface, resulting in a portion of free patellar tendon in the femoral tunnel immediately proximal to the articular surface (Fig. 1). This shifted the constructs proximally compared to the arrangement commonly used by surgeons in which the distal end of the femoral bone block is aligned with the articular margin of the femoral tunnel. As a result tunnel enlargement in the PT group was expected to be more apparent on femoral side than on the tibial side.

In both groups low suction drains were inserted into the joint and into the subcutaneous layer. The patients were discharged

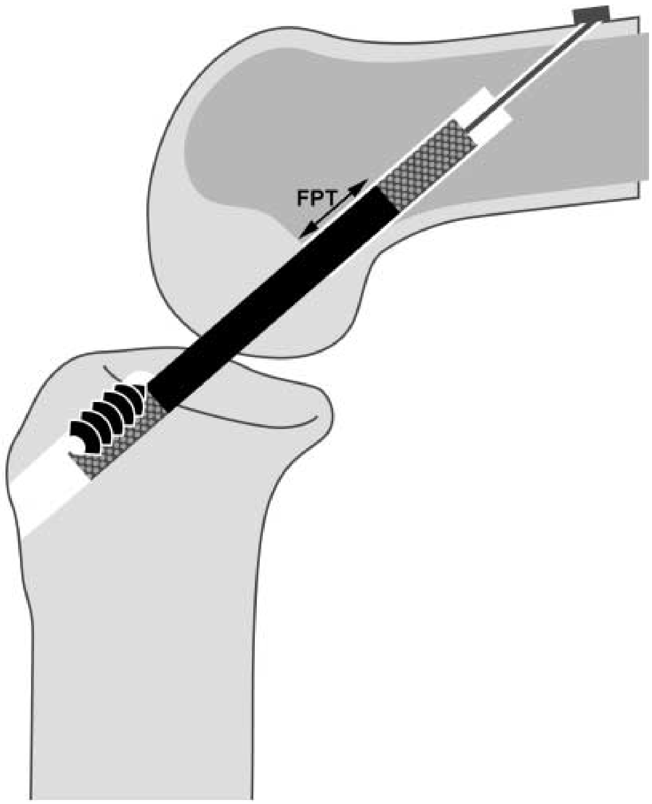

Fig. 1 Schematic diagram of the patellar tendon graft construct demonstrating the free patellar tendon (*FPT*) in the femoral tunnel and the absence of free patellar tendon in the tibial tunnel

from hospital 24 to 48 h after surgery. All patients in both groups followed the same accelerated rehabilitation protocol [21]. This emphasised immediate restoration of full extension and quadriceps function and allowed full weight bearing as tolerated from the first postoperative day. Braces were not used in either group.

Evaluation

Patients were studied radiographically at 4 months and 1 and 2 years postoperatively. A flexion weight bearing anteroposterior (AP) view and a lateral view in full extension were obtained. Tunnel enlargement was studied according to L'Insalata et al. [10]. Using a digital caliper the distance between the sclerotic margins of each tunnel was measured at its widest dimension, together with the diameter of the fixation post head or interference screw. This was completed independently by two observers. The size of the drill bit used for tunnel reaming was recorded at the time of surgery, and the diameters of the fixation post head and interference screw were known and constant. All measurements were corrected for magnification by comparing the measured and real diameters of the interference screws and fixation post heads. This was performed separately for AP and lateral views. Change in tunnel size was calculated as a percentage of the diameter of the drill bit used. In all, 58 radiographs were available for measurement at 4 months, 54 at 1 year and 51 at 2 years. To enable a determination of inter-observer variability, ten radiographs were randomly sampled and reviewed by both observers and a third examiner.

The position of the tibial tunnel relative to Blumensaat's line was also recorded on the most recent lateral radiograph for each patient using a method similar to that described by Muneta et al. [12]. The anterior margin of the tibial tunnel was rated as being anterior to, in line with, or posterior to Blumensaat's line. In those patients in the PT group in whom no tunnel margin could be identified, the anterior margin of the interference screw was used. This screw was always inserted anterior to the distal bone block.

The shape of the tunnels was classified into one of three descriptive categories according the system described by Peyrache et al. [19]. This describes widening as being either cone type, line type or cavity type.

Each review also consisted of a clinical evaluation including measurement of extension deficit by the method of Sachs et al. [20], KT-1000 measurement of side to side differences in anterior tibial translation at 15 lb (67 N) and 30 lb (134 N), and recording of the International Knee Documentation Committee (IKDC) score and Cincinnati Knee Score [15].

Statistical analysis

The percentage of tunnel change was compared between the HS and PT groups using an independent samples *t* test. Analysis of variance was used to compare tunnel change over time, and was performed separately for each group. Intra-class correlation coefficients were performed to determine inter-observer reliability.

To examine whether radiographic change is related to clinical outcome, we determined the correlation between the percentage of tunnel change and the following variables: extension deficit, KT-1000, IKDC score, and Cincinnati Knee Score. For this analysis data from the 2-year follow-up was used. The original scores were converted to ranks, and the relationship of the ranks was measured separately for each group using Spearman's correlation. This method of analysis was chosen because we wanted to measure the consistency of the relationship between tunnel enlargement and clinical outcome independent of its specific form (i.e. we did not necessarily expect relationships to be linear).

Results

Radiographic results

Because of the presence of the interference screw and the proximity of the bone block to the tibial articular surface the tibial tunnel could not be reliably measured in the PT group. The distribution of tunnel enlargement in the two groups is shown in Table 2. In the PT group nine patients (32%) had femoral tunnel obliteration (on both AP and lateral views) from 4 months onwards. In the HS group well-defined sclerotic margins were always present for

Table 2 Incidence of bone tunnel widening; results are based of the average of all follow-up data and lateral and AP views

	Obliterated		<0%		0–25%		26–50%		>50%	
	n	%	*n*	%	*n*	%	*n*	%	*n*	%
Femur										
Hamstring	–	–	–	–	2	6	21	64	10	30
Patellar tendon	9	32	3	11	13	46	2	7	1	4
Tibia										
Hamstring	–	–	–	–	22	67	9	27	2	6

both the tibial and femoral tunnels. No patient in the HS group showed femoral tunnel obliteration similar to that seen in the nine PT patients.

Radiographic tunnel widths were significantly larger than the size of the femoral and tibial tunnels drilled at surgery for the HS group at each follow-up (P<0.001). In the PT group the radiographic width of the femoral tunnel was significantly larger than the size of the tunnel drilled at surgery for the AP view (P<0.05 at each follow-up) but not the lateral view. Radiographic tunnel width did not significantly change between 4 month and 2 years for either group. These data, and the comparisons detailed below, exclude the nine PT patients in whom the femoral tunnel appeared obliterated.

The mean percentage changes in tunnel width on each radiographic view and at each follow-up for both groups are presented in Table 3. The HS group showed significantly greater femoral tunnel widening than the PT group at all follow-up times on both AP and lateral views.

All patients in the HS group showed predominantly linear tunnel expansion in both the femoral and tibial tunnels. All PT patients in whom the femoral tunnel was visualised showed linear tunnel expansion, except for one patient with marked tunnel enlargement in a cone-shaped pattern.

Inter-observer variability for tunnel width measurement was satisfactory. The average intraclass correlation coefficient was 0.94 with a 95% confidence interval of 0.88 to 0.97.

On the most recent lateral radiograph with knee in full extension the anterior margin of the tibial tunnel was in line with or posterior to Blumensaat's line for all patients in both groups, except for one patient in the PT group who also showed cone shaped femoral and tibial tunnel widening (PT: 1 anterior, 5 in line, 22 posterior; HS: 3 in line, 30 posterior).

Table 3 Percentage tunnel widening in hamstring and patellar tendon groups after 4 months, 1 and 2 years postoperatively

	4 months	1 year	2 years
Femur			
Anteroposterior view			
Hamstring	49.5±19.8**	47.9±18.8**	47.4±18.3**
Patellar tendon	16.2±17.4	15.8±21.1	15.6±17.4
Lateral view			
Hamstring	42.8±18.5**	36.3±18.6*	35.9±16.3*
Patellar tendon	11.3±23.9	11.9±22.4	10.5±26.6
Tibia			
Anteroposterior view			
Hamstring	23.1±13.2	24.3±15.9	24.1±16.6
Patellar tendon	–	–	–
Lateral view			
Hamstring	23.3±19.6	21.8±14.3	21.8±18.0
Patellar tendon	–	–	–

*P<0.001, **P<0.0001 HS vs. PT

Clinical outcomes

The clinical assessment results at 2 years are summarised in Table 4. Overall, a KT-1000 side-to-side difference greater than 3 mm was found in only one patient (HS) at 15 lb and in only five patients (three HS, two PT) at 30 lb. The majority of patients in both groups had an extension deficit less than 3°. Four patients scored a 'severely abnormal' IKDC rating, and in all four cases this was due to

Table 4 Clinical results from the 2-year review; 2-year data was unavailable for seven patients (five Hamstring, two patellar tendon)

	Hamstring		Patellar tendon	
	n	%	*n*	%
KT-1000 side to side difference (mm)[a]				
15 lb				
0	5	16	6	26
>0–2	14	45	15	65
>2–3	11	36	2	9
>3–5	1	3	–	–
Average (mm)**	1.4		0.7	
Range (mm)	0 to 3.5		–1 to 2.5	
30 lb				
0	4	13	7	30
>0–2	17	55	6	26
>2–3	7	23	8	35
>3–5	3	9	2	9
Average (mm)	1.7		1.1	
Range (mm)	0 to 4		–1 to 3.5	
Extension deficit (degrees)[a]				
0	9	29	2	9
>0–3	17	55	11	48
>3–5	4	13	6	26
>5–10	1	3	4	17
Average*	–1.7°		–3.1°	
Range	–8 to 0		–9 to 0	
IKDC score				
A	8	26	6	26
B	11	35	8	35
C	10	32	7	30
D	2	7	2	9
Cincinnati Knee Score				
≤50	1	3	–	–
>50–80	2	7	5	22
>80–100	15	48	10	43
100	13	42	8	35

*P<0.05, **P<0.01 HS vs. PT
[a]Operated knee compared to contralateral knee

a poor score in the symptoms category. Two of these patients were not undertaking moderate or strenuous activities (one due to pregnancy and one due to ceasing sporting activities). Cincinnati Knee Score values greater than 80 were obtained in 85% of patients.

Clinical correlations

There was no correlation between clinical variables and tunnel enlargement.

Discussion

The results of the present study demonstrate femoral bone tunnel enlargement following ACL reconstruction to be more common and greater with HS grafts than with PT grafts. Such results are consistent with previous reports by Clatworthy et al. [3] and L'Insalata et al. [10]. When considered together, the three studies indicate bone tunnel enlargement to be a common occurrence of HS graft ACL reconstruction. A notable finding of the present study was the complete obliteration of the femoral tunnel seen in 32% of PT patients. No HS patient showed similar tunnel obliteration. This highlights the marked differences in the pattern of bone tunnel change between the two graft types.

In the past, grafts fixed by suspensory methods have been said to be susceptible to longitudinal motion due to distance between the graft and the point of fixation, the so-called 'bungy effect' [7]. This increased potential for graft motion has been used to explain femoral tunnel enlargement with HS grafts [8, 10, 13]. However, in the present study only 11% of PT patients had femoral tunnel widening greater than 25%, compared with 94% of HS patients, despite the bungy effect being potentially present in both groups. Clatworthy et al. [3] reported similar findings in relation to PT and HS grafts and in a more recent study, comparing different fixation methods for HS grafts only, reported the least amount of tunnel enlargement with Endobutton fixation compared to other fixation methods [2].

Although femoral suspensory fixation may not be biomechanically as sound as other methods, it alone does not in itself appear to be the principal cause of femoral tunnel enlargement, at least for PT grafts. For HS grafts the situation is less clear because in this study suspensory fixation was used on the tibial side as well. If metallic interference screw fixation had been used, the two groups would have had identical fixation methods, which would have allowed more definite conclusions. Tibial interference screw fixation of the HS grafts was not used in this study because the clinical efficacy of this method had not been clearly established at the time the study was conceived.

In this study a difference between the two groups was the content of the femoral tunnel. In the PT group there was both free patellar tendon and bone compared to the HS group where there was free tendon only. It may be that it is the presence of what is effectively a bone graft that reduces the incidence and extent of tunnel enlargement with PT grafts. Morris et al. [11] recently reported the results of a sheep model in which extensor tendon grafts were implanted into bone tunnels with and without recombinant human bone morphogenetic protein 7. In the animals in the bone morphogenetic protein 7 group, there was increased bone formation at the tendon-bone interface as well as new bone formation in the tendon itself from 6 weeks after implantation. We postulate that the PT bone block behaves in a similar manner, particularly because the presence of free tendon in the femoral tunnel in the PT group did not appear to mitigate against tunnel obliteration. This is further supported by the recent study by Peterson and Laprell [18]. They reported fibrocartilage healing at the tendon-bone interface in the femoral tunnel of PT autografts with both interference screw fixation and suspensory fixation compared to fibrous healing for HS autografts with suspensory fixation.

In the present study we were unable to measure the tibial bone tunnel in the PT group. This is comparable to the inability or difficulty in measuring the femoral tunnel in the PT group in previous studies [3, 10, 19] in which the graft constructs were inserted such that distal end of the proximal bone block was flush with the articular surface of the femur. Use of the Endobutton allowed us to shift the PT constructs proximally, thereby exposing free patellar tendon in the femoral tunnel and thus allowing a situation more similar to that of the HS grafts.

The present data also clearly demonstrate tunnel enlargement at 4 months that does not significantly change at 1 and 2 years. This is the first study to describe the course of bone tunnel enlargement over time for both HS and PT grafts, although previous studies showed that tunnel enlargement does not change from 3 months to 2 years in patients treated with PT grafts [4, 19]. It appears that bone tunnel enlargement develops during the first few months after surgery, with little change thereafter. Accelerated rehabilitation programmes have been proposed as a contributing factor for bone tunnel enlargement, and it has therefore been suggested that it may be necessary to proceed more slowly with rehabilitation following HS ACL reconstruction than with PT reconstruction [3, 8]. If slower rehabilitation were to have an effect it would clearly need to be implemented during the first 3 months.

As in previous studies, we did not observe a clinically apparent effect of tunnel enlargement in the short term [3, 5, 8, 9, 13, 19]. Although there were differences between the two groups in terms of anterior knee laxity and extension deficit, there was no correlation between these or other variables and tunnel enlargement. Despite the greater incidence and extent of extension deficit in the PT group, the magnitude of the difference was small and was not related to tibial tunnel position, which was assessed as being satisfactory in all but one patient. Peyrache et al. [19] observed that cone-shaped enlargement was associated with a trend to increased anterior tibial laxity but did not find any correlation between tunnel enlargement and tibial tunnel position. Long-term follow-up is, however, required to clearly establish any clinical effects of tunnel enlargement, including any implications for revision surgery. Enlarged tunnels may potentially cause problems with graft positioning and fixation in this situation.

It is interesting that the only PT patient to show relatively marked tunnel enlargement (>50% cone shaped) was also the same patient in whom the anterior margin of the tibial tunnel was anterior to Blumensaat's line. Clinical data from this patient showed an extension deficit of 3.3° but no KT-1000 side-to-side difference in anterior tibial translation at 30 lb. We make no attempt, however, to draw conclusions based on only one subject but simply note the observation.

Conclusions

In this study femoral bone tunnel enlargement following ACL reconstruction was shown to be more frequent and greater with hamstring grafts than in patellar tendon grafts. The apparent femoral tunnel obliteration in 32% of patellar tendon patients suggests a fundamental difference between the graft constructs. Because of the different tibial fixation methods used in this study we cannot be certain that suspensory fixation does not play a role in tunnel enlargement. However, suspensory fixation was used on the femoral side for both graft types, and this may point to a biological contribution to the observed differences. The patellar tendon bone block may constitute such a biological factor. It is possible that the presence of what is effectively a bone graft may reduce bone tunnel enlargement. Whatever its cause, in the short-term bone tunnel enlargement appears to have no effect on the clinical outcome of ACL reconstruction surgery.

References

1. Aglietti P, Zaccherotti G, Simeone AJ, Buzzi R (1998) Anatomic versus non-anatomic tibial fixation in anterior cruciate ligament reconstruction with bone-patellar tendon-bone graft. Knee Surg Sports Traumatol Arthrosc 6 [Suppl 1]:S43–S48
2. Clatworthy M, Bartlett J, Howell S, L'Insalata J, Fu F, Harner C (1999) The effect of graft fixation techniques on tunnel widening in hamstring ACL reconstruction. Arthroscopy 15:S5
3. Clatworthy MG, Annear P, Bulow JU, Bartlett RJ (1999) Tunnel widening in anterior cruciate ligament reconstruction: a prospective evaluation of hamstring and patella tendon grafts. Knee Surg Sports Traumatol Arthrosc 7: 138–145
4. Dyer CR, Elroy BF (1995) Tibial and femoral bone tunnel enlargement following allograft replacement of the anterior cruciate ligament. Arthroscopy 11:353–356
5. Fahey M, Indelicato PA (1994) Bone tunnel enlargement after anterior cruciate ligament replacement. Am J Sports Med 22:410–414
6. Hoher J, Moller HD, Fu FH (1998) Bone tunnel enlargement after anterior cruciate ligament reconstruction: fact or fiction? Knee Surg Sports Traumatol Arthrosc 6:231–240
7. Hoher J, Livesay GA, Ma CB, Withrow JD, Fu FH, Woo SL (1999) Hamstring graft motion in the femoral bone tunnel when using titanium button/polyester tape fixation. Knee Surg Sports Traumatol Arthrosc 7: 215–219
8. Jansson KA, Harilainen A, Sandelin J, Karjalainen PT, Aronen HJ, Tallroth K (1999) Bone tunnel enlargement after anterior cruciate ligament reconstruction with the hamstring autograft and endobutton fixation technique. A clinical, radiographic and magnetic resonance imaging study with 2 years follow-up. Knee Surg Sports Traumatol Arthrosc 7:290–295
9. Linn RM, Fischer DA, Smith JP, Burstein DB, Quick DC (1993) Achilles tendon allograft reconstruction of the anterior cruciate ligament-deficient knee. Am J Sports Med 21: 825–831
10. L'Insalata JC, Klatt B, Fu FH, Harner CD (1997) Tunnel expansion following anterior cruciate ligament reconstruction: a comparison of hamstring and patellar tendon autografts. Knee Surg Sports Traumatol Arthrosc 5:234–238
11. Morris H, Nicklin S, Harrison J, Yu Y, Walsh WR (2000) Augmentation of tendon-bone healing with exogenous OP-1 in an ovine ACL reconstruction Presented at the ACL Study Group, Rhodes, Greece
12. Muneta T, Yamamoto H, Ishibashi T, Asahina S, Murakami S, Furuya K (1995) The effects of tibial tunnel placement and roofplasty on reconstructed anterior cruciate ligament knees. Arthroscopy 11:57–62
13. Nebelung W, Becker R, Merkel M, Ropke M (1998) Bone tunnel enlargement after anterior cruciate ligament reconstruction with semitendinosus tendon using Endobutton fixation on the femoral side. Arthroscopy 14:810–815
14. Noyes FR, Stabler CL (1989) A system for grading articular cartilage lesions at arthroscopy. Am J Sports Med 17:505–513
15. Noyes FR, McGinniss GH, Mooar LA (1984) Functional disability in the anterior cruciate insufficient knee syndrome. Review of knee rating systems and projected risk factors in determining treatment. Sports Med 1:278–302
16. Noyes FR, Barber SD, Mooar LA (1989) A rationale for assessing sports activity levels and limitations in knee disorders. Clin Orthop 246:238–249
17. Noyes FR, Mooar LA, Barber SD (1991) The assessment of work-related activities and limitations in knee disorders. Am J Sports Med 19:178–188
18. Petersen W, Laprell H (2000) Insertion of autologous tendon grafts to the bone: a histological and immunohistochemical study of hamstring and patellar tendon grafts. Knee Surg Sports Traumatol Arthrosc 8:26–31
19. Peyrache MD, Djian P, Christel P, Witvoet J (1996) Tibial tunnel enlargement after anterior cruciate ligament reconstruction by autogenous bone-patellar tendon-bone graft. Knee Surg Sports Traumatol Arthrosc 4:2–8
20. Sachs RA, Daniel DM, Stone ML, Garfein RF (1989) Patellofemoral problems after anterior cruciate ligament reconstruction. Am J Sports Med 17:760–765
21. Shelbourne KD, Nitz P (1990) Accelerated rehabilitation after anterior cruciate ligament reconstruction. Am J Sports Med 18:292–299